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DDoS Attack Detection Model using Machine Learning Algorithm in Next Generation Firewall

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Abstract

Next generation firewall is taking major part to secure network environment in the industry. This device will monitor all the traffic which is coming inside the network or going outside of the network. With all these security devices attackers can still perform various kind of attacks on the network. DDoS attack is one of the hardest attack to identify which will send packets to the network and which will look like normal traffic but it will act as a DDoS traffic. In this paper we used binary decision tree, XGBoost and support vector machine to identify DDoS attack traffic pattern from the different features in the packet header. Data will be fetched from the packet header and among them standard deviation of the packet bytes and packet flows are the features considered. We have applied this data on the trained dataset. Algorithm will predict whether the traffic coming inside the network is trusted or not. Out of the three algorithms Binary Decision tree algorithm is giving 99 percentage of accuracy and will predict the data as fast as possible. Here priority is to filter DDos attacks of any security level in the line speed of the NIDS or any other appliances. This method of DDoS attack detection will add extra layer of security in the next generation firewall which will make firewall more robust.

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Keywords: Next Generation Firewall;Distributed Denial-of-Service Attack;Machine Learning Algorithm

1. Introduction

Next generation firewall(NGF) is widely used in today's network and it provides more security compared to traditional firewall as it will improve traffic inspection. Traditional firewall will work on port, protocol and the rules defined on the firewall. On the other hand, next generation firewall will do all the same jobs as the traditional firewall and also do the layer 7 inspection and will defend the network against the new and advance threats. Few of its best feature is web-filter, application control, Secure Sockets Layer (SSL) inspection and IPS/IDS/AV (antivirus) and distributed

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denial-of-service (DDOS). Web-filter is widely and most often used feature in the next generation firewall. With the help of web-filter the administrator can allow or deny user based on the URL. Since it supports patterns like simple, wild card and regex, we can configure URL restriction with the help of this. Web-filter works on application layer and will monitor the SSL connection initiated by the user and then as per the policy configured, the traffic will be allowed or denied.
Application control is similar to the web-filter and it requires SSL inspection to work further. So when few packets from client to server and server to client is analysed based on the traffic signature the firewall will identify the access of the application user. If user is using custom application, then also we can add that signature in the application control and control the traffic with it. SSL inspection will allow the users to inspect the HTTPS traffic between client and server, it will work like man-in-middle. When the user initiate the traffic then for the destination the server certificate will be replaced by firewall's own certificate and the firewall will provide its certificate to the user. Now the firewall will initiate the connection with the server on behalf of the user and due to that it can decrypt the traffic and apply the profiles configured for the traffic. Next generation firewall is also capable for performing the activities for Intrusion detection or prevention systems (IPS/IDS) and anti-virus. Administrator can create policy for the non-trusted traffic and when the traffic reaches to the device it will scan its database for the patterns and the signature. If its matches then it will apply the action configured for the same. Next generation firewall also supports sandboxing, so if a new signature is being identified by the device then it will sent the file to sandboxing and there it will be scanned and a signature will be provided to all the device in the next update.

1.1. Motivation for this work

DDoS attack traffic almost look similar as normal network traffic, so it is very hard to classify the packet is normal network packet or it is DDoS attack packet. Next generation firewall is already having deep packet inspection feature which is also not fully accurate and is still not able to identify all the DDoS attack parameters and behaviour of the attack. Packet classification [1, 2, 3] is very important step in next generation firewall, in which network packet will get classify for the proper application or security. Hence we are addressing the same in this paper.

1.2. Approach

When any network packet try to enter in industrial network it should first cross the next generation firewall. When firewall analyse the packet, it will check all the rules which is define by the network administrator. DDoS attack traffic almost look like normal traffic. In this work we used machine learning algorithm which will identify the DDoS attack traffic in the normal packet flowing with use of standard deviation of flow and packets. It will continuously monitor network traffic and try to identify the DDoS attack traffic. If it finds any packets which trigger DDoS attack it will generate alert and send notification to the network administrator.

2. Literature Survey

Some of the work done in this area are summarised in this section. Partha Chakraborty et al., [4] in his work added artificial intelligence with firewall which will make firewall adept of manipulating traffic behavior by making use of incoming traffic and dynamically creating some rules with itself for some unique packets. Hence no default measures were used. Neupane, Kishan et al., [5] have briefed the importance of the next generation firewalls when compared to the traditional firewalls. In [6] the system includes determining an application layer signalling, determining requirements of a service media flow on security level and performing a security inspection of a signalling flow and determining control of the service media flow on security level according to a stored policy and the requirements of the service media flow on security level. Conventional firewalls are finding difficult to capture the evolving threats which force the industries and organisations to deploy the NGF which will offer higher efficiency, better performance, high protection, and simplicity. [11, 12] papers discuss about the need for NGF, its importance, and the industrial implementation trends. Also gives an idea of how NGF can support AI/ML algorithms and hence improve protection to IoT systems.

In literature multiple-producer and multiple-consumer queue defined by Tsigas and Zhang [9] , Devassy, Rahul, et al [10] , Ni, Shih-Hsiung [13] all improved on former performance results keeping the need for expensive hardware

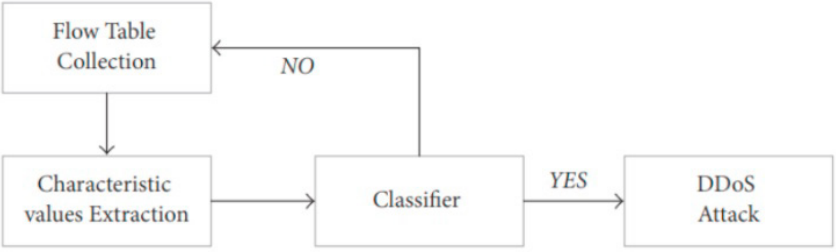


Fig. 1: DDoS Attack Classification Model

synchronization primitives. Benfano Soewito et al., [16] analyze the usefulness of the NGF that can be implemented to secure IoT networks. They have experimented the DDoS and SQL Injection on all networks. There has been previous research done to commonly analyze the effectiveness of network security and NGF by concentrating on how to evade attacks. Packet queuing concept and pipelined packet switching has been studied in [7, 8, 14, 15]. Network intrusion detection system can be used for detection and classification of network breaches by using shallow and deep networks[19, 21]. This method can be further incorporated as a module in NGF. A novel method has been experimented in [20] by adding role based access control in the firewall rule which will be represented in a three-dimensional matrix and authors implemented this work over a distributed firewall. Authors in [22] included methods to improve detection using the feature of intrusion prevention system in firewall by considering the packet flows only. Their motivation is to detect the attack, analyze it, and drop the attacked packet if it is risky before it harms and alert the network administrator. In [18] the authors expalins why we need to adopt the NFG and how it offers good performance, protection, and better simplicity. Paper also explains the need for NGFW, its features, and the industrial adoption trends. They also justifies how NGFW can integrate the promising machine learning algorithms and better safeguard IoT systems. In existing literature of DDoS attack detection authors does not mention the change of the speed of source port in attack detection when extracting the traffic characteristic values, and a large number of new port addresses were randomly generated in the attack process. We address some of these issues in this paper.

3. Proposed Model

Next Generation firewall is the most important device to secure the network environment. This should be able to protect from the all the modern attacks in the environment. DDoS attack is most powerful attack which is also very hard to detect since it act as a normal traffic in the environment. For firewall also it will act as a legitimate traffic so it is very hard to detect the DDoS attack. In this work we proposed binary decision tree based packet classification model which will predict the traffic is normal traffic or the DDoS attack traffic. This model will give an alert to the administrator when a packet is identified as a DDoS attack packet. Finer classification within the DDos attacks is out of reference in this paper. Here the priority is to filter DDos attacks (of any security level) in the line speed of the NIDS, appliances.

We have created Binary Decision Tree Based Packet Queuing Schema for Next Generation Firewall [23] and have implemented DDos attack in this framework. This paper focus on detecting DDoS attack using binary decision tree, XGBoost and support vector machine algorithm.

The flow diagram in figure 1 consists mainly of the flow state collection, the extraction characteristic values, and the classifier judgment. The flow state collection periodically sends a flow table request to the Openflow switch and sends the flow table information replied from the switch to the flow state collection. The network packet will add to the flow table collection and this table data will send it to the feature extraction. In stage of feature extraction, we are taking some of the parameters from the packet header which will help us to identify the DDoS attack traffic from the normal packet flow. These extracted features are giving information which are used for classifying with machine learning algorithm and hence to identify the DDoS attack traffic from the normal traffic. When DDoS attack start in the network environment, this program will send an alert to the administrator. In DDoS attack, the network will flood

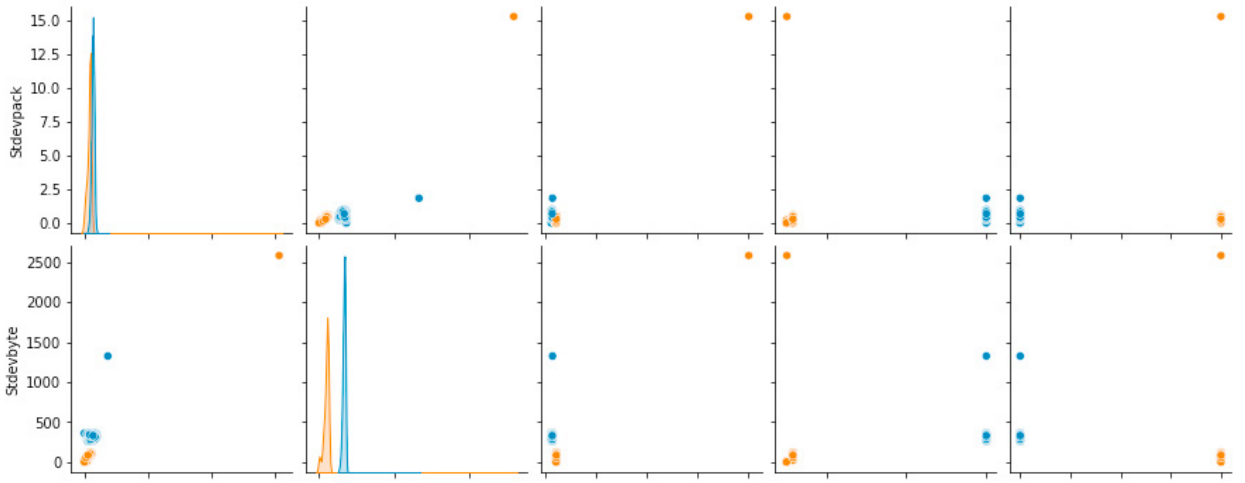


Fig. 2: DDoS Attack Classification Model

with large number of packets and so packet flow bytes and packet flow speed will increase. The proposed model will work on the statistic of the standard deviation of flow byte and standard deviation of flow packet. These two tuples of features will get affect when DDoS attack is happening in the network.

3.1. Extracted features

The following two-tuple characteristic values related to DDoS attacks are used for DDoS attack detection.

1. Standard Deviation of Flow Packets (SDFP): The standard deviation of the number of packets in the T period is calculated as follows:
   $\text{SDFP} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(Packets_i - MeanPackets)^2}$
   where $Meanpackets = (1/N)\sum_{i=1}^{N} Packets_i$ represent the average number of the packets in the T period, N is the total number of flow entries per period. During the event of an attack, in order to produce the attack effect, the general attack data packets are relatively small and the standard deviation of flow packets will be smaller than the normal flow.

2. Standard deviation of Flow Bytes (SDFB): The standard deviation of the number of bits in the T period, is as follows:

   $\text{SDFP} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(Byte_i - MeanBytes)^2}$
   where $MeanBytes = (1/N)\sum_{i=1}^{N} Byte_i$
   represent the average of the number of bits in the T period. During the event of an attack, in order to reduce the packet load, attacker will send a smaller bit of data packets and the standard deviation flow bits will be smaller than the normal flow.

While selecting the two fields from the IP header we have done analysis between standard deviation of flow packets and standard deviation of flow bytes as seen in the figure 2. we have found that these two features are getting affected when the DDoS is performed in the network. Standard deviation of flow byte number is increasing when attack is happening in the environment and Standard deviation of flow packet is almost zero while attack is going on. Standard deviation of flow packet is dependent on response of the sender packet but this is not possible in DDoS attack because in this attack receiver is getting large number of packet so receiver won't respond to every packet. That's why it reduce to the zero in graph. In this proposed model we calculate standard deviation of flow byte and standard deviation of

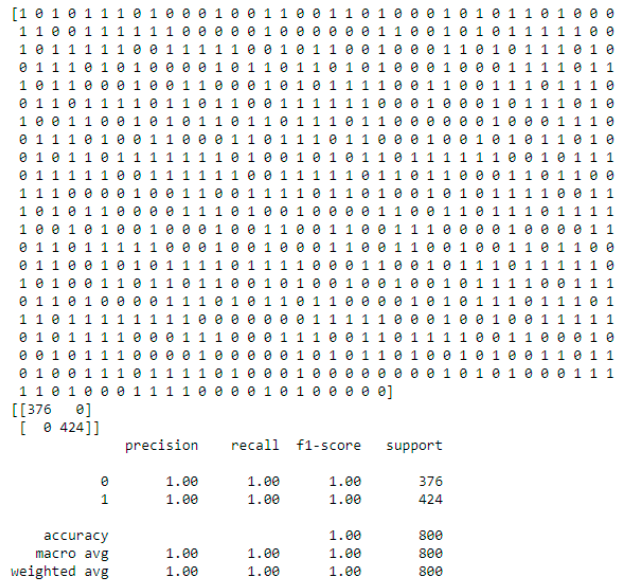


Fig. 3: Result of Binary decision tree algorithm based on experiment-I

flow packet of three combine packet. Final stage we use testing dataset to verify the validated data set and check whether real time data will get differentiated between the DDoS attack traffic or the normal traffic.

4. Experiments and Results

We downloaded one network packet dataset from the internet which will work as a training dataset as a packet will come to the firewall. Every packet will get predicted from this training dataset. We captured wireshark log details for creating testing dataset. We captured logs of normal traffic and DDoS attack traffic through wireshark and have stored that data in the excel file.

4.1. Experiment-I

During this experiment we distributed the data into the two section which are testing (20%) and training dataset (80%). The model is trained with 80% of trained dataset and then validated with the testing data by predicting the DDos attack from the different IP header parameters. We have used binary decision tree, XGBoost and support vector machine algorithm for predicting the DDoS attack. Among the three algorithms binary decision tree is more effective and efficient. As shown in the figure 3, we can summarise that from the 800 packets program predicted 424 DDoS attack traffic and 376 normal packets. When it comes to the accuracy part, confusion matrix is also giving us 100 percent accuracy rate. When comparing the result of XGboost and SVM algorithm both of them is also showing same accuracy but the efficiency is different as seen in figure 4. We can conclude that Binary decision tree algorithm is given more accuracy in very less time period.

4.2. Experiment-II

We have captured live network packet through wireshark while doing DDoS attack through kali linux tool hping3 and normal web surfing. We stored that dataset into the excel file and used that as a testing dataset. We applied this real time testing dataset to the trained model of binary decision tree, XGboost algorithm and SVM. In this experiment binary decision tree algorithm was giving a prediction accuracy 99 percentage and in figure 5 we observed that total 11,92,432 packets predicted from that total 1168974 normal traffic and 23458 DDoS attack traffic predicted. Same

```
[1 1 0 0 0 0 0 1 1 0 0 0 0 0 0 1 1 0 0 1 1 1 1 0 0 0 0 1 0 1 0 1 0 0 1 0 0
 1 1 0 0 1 0 1 0 0 0 1 1 1 1 1 1 0 0 1 0 1 0 1 1 1 1 1 0 1 0 0 0 1 1 1 1 0 1
 0 1 1 1 0 1 0 0 0 0 1 0 1 0 0 1 0 1 0 1 0 0 0 1 0 0 1 0 1 1 1 0 1 0 0 0 1
 0 1 1 0 1 0 1 1 0 1 1 0 1 1 0 1 1 0 0 1 0 1 1 1 1 1 1 0 0 1 1 1 1 1 0 1 0 1 0
 1 1 0 0 1 0 1 1 0 1 0 0 0 1 0 0 1 0 1 0 0 1 1 0 1 0 1 0 1 0 0 0 1 0 0 0 0
 1 0 0 1 1 0 0 1 1 0 0 1 1 1 1 0 1 1 1 1 1 1 1 0 1 1 0 0 1 0 1 0 0 1 0 1 0 1
 1 1 1 0 0 1 0 1 0 0 1 0 1 1 0 0 0 0 0 0 0 1 1 1 1 1 0 1 1 0 0 1 0 1 1 0 1
 1 1 1 0 1 1 0 0 1 1 1 0 0 0 1 1 0 1 0 0 0 0 0 1 0 0 0 0 1 0 1 1 0 0 0 0 1
 1 1 0 1 1 1 0 1 1 0 0 0 0 1 0 1 0 0 1 0 0 1 1 0 0 1 1 0 1 0 1 0 1 1 1 1 0 0 1 1 0 0
 1 0 1 1 0 1 0 0 1 1 1 0 0 1 1 0 1 0 1 1 0 0 1 1 0 1 0 1 0 0 0 0 0 1 0 0 1
 0 0 0 1 0 1 1 1 1 1 0 1 1 0 1 1 0 1 0 0 0 0 0 0 0 1 0 0 0 1 0 0 1 0 0 0 0
 1 1 1 1 0 0 1 1 0 0 1 1 1 0 1 0 1 1 1 1 1 0 0 1 0 1 0 1 0 1 0 0 0 0 1 0 0
 1 1 1 0 1 1 1 1 0 1 1 1 0 0 0 0 0 1 1 1 0 0 1 1 1 0 1 0 0 0 1 0 1 0 0 0 1
 0 1 1 1 0 0 1 1 0 0 1 1 0 0 0 1 0 1 1 0 0 1 0 0 1 1 1 1 0 1 0 0 0 1 0 0 0
 0 1 0 1 0 0 0 0 0 0 1 0 0 0 0 0 1 1 0 0 1 1 0 0 1 0 1 1 0 0 1 0 1 1 0 0 1
 1 1 1 1 0 0 1 0 0 1 1 0 1 1 1 1 1 0 1 0 1 1 0 1 0 0 1 0 1 0 0 0 1 1 1 1 1
 1 1 0 0 0 1 1 0 1 1 0 1 0 0 1 0 0 1 0 1 0 1 1 1 0 1 1 1 1 0 1 1 1 0 0 0 1
 1 0 0 1 0 0 1 0 1 0 1 0 1 1 1 0 0 1 1 1 0 0 0 1 1 1 1 0 0 1 0 0 1 1 1 0 0
 1 0 0 1 0 1 0 0 0 1 0 0 1 1 1 0 0 0 0 1 1 1 0 1 0 0 0 0 1 0 0 1 1 1 0 1 1
 1 1 0 1 1 0 0 0 0 1 0 0 0 0 1 1 0 1 1 1 0 0 1 0 1 0 1 0 1 1 0 1 0 0 0 1 1
 0 1 0 1 1 0 0 1 0 0 0 1 0 1 0 1 1 0 0 1 0 0 0 0 1 0 0 0 0 1 0 1 0 0 1 1 1
 0 1 0 0 0 1 0 0 0 0 0 1 1 0 1 1 0 0 1 1 1 1 1 1]
[[407   0]
 [  0 393]]
              precision    recall  f1-score   support

           0       1.00      1.00      1.00       407
           1       1.00      1.00      1.00       393

    accuracy                           1.00       800
   macro avg       1.00      1.00      1.00       800
weighted avg       1.00      1.00      1.00       800
```

Fig. 4: Result of XGboost algorithm based on experiment-I

```
[[   6043   17415]
 [      0 1168974]]
              precision    recall  f1-score   support

           0       1.00      0.26      0.41     23458
           1       0.99      1.00      0.99   1168974

    accuracy                           0.99   1192432
   macro avg       0.99      0.63      0.70   1192432
weighted avg       0.99      0.99      0.98   1192432
```

Fig. 5: Result of Binary Decision Tree Algorithm based on experiment-II

testing data was given to trained model of XGboost algorithm and SVM. After applying testing data set on both the model accuracy remains 99 % for XGboost and 96 % for SVM (figure 6).

4.3. Time Efficiency

In this proposed model we have focused on the parameter time efficiency which most of literature work has not considered. We have compared the time efficiency of all three machine learning algorithms and have calculated the time for every one prediction. As shown in the figure 7, Binary decision tree is taking lesser time as compare to the other two machine learning algorithms. It is taking on average of 1.1 millisecond or 0.001 seconds for one packet prediction. Hence compared to the other algorithm this is the fastest. When we checked using the XGBoost Algorithm it is taking on average 7.5 millisecond(figure 8) which is equal to 0.007 seconds and SVM takes 0.001 seconds only. Table 1 summarizes the results in terms of accuracy and efficiency for three algorithms used for DDoS attack detection.

```
[0 1 1 ... 1 1 1]
[[   6185   17273]
 [      0 1168974]]
              precision    recall  f1-score   support

           0       1.00      0.26      0.42     23458
           1       0.99      1.00      0.99   1168974

    accuracy                           0.99   1192432
   macro avg       0.99      0.63      0.70   1192432
weighted avg       0.99      0.99      0.98   1192432
```

Fig. 6: Result of XGboost Algorithm based on experiment-II

```
time==> 1.1406999983591959 milliseconds
time==> 1.0951000003842637 milliseconds
time==> 1.1301000049570575 milliseconds
time==> 1.0939999992842786 milliseconds
time==> 1.1130999992019497 milliseconds
time==> 1.093499995477032 milliseconds
time==> 1.1150999998790212 milliseconds
time==> 1.1255999997956678 milliseconds
time==> 1.0971000010613352 milliseconds
time==> 1.1694999993778765 milliseconds
time==> 1.1356000031810254 milliseconds
time==> 1.1133000007248484 milliseconds
time==> 1.1005999986082315 milliseconds
time==> 1.0977000056300312 milliseconds
time==> 1.0959999999613501 milliseconds
time==> 1.095200001145713 milliseconds
time==> 1.109699995140545 milliseconds
time==> 1.1804000023403205 milliseconds
time==> 1.1427999997977167 milliseconds
```

Fig. 7: Execution time of Binary Decision Tree Algorithm

```
time==> 6.333600002108142 milliseconds
time==> 5.925099991145544 milliseconds
time==> 6.569299992406741 milliseconds
time==> 7.870400004321709 milliseconds
time==> 4.811299993889406 milliseconds
time==> 5.04420000652317 milliseconds
time==> 5.646699995850213 milliseconds
time==> 7.43030000012368 milliseconds
time==> 7.0324999978765845 milliseconds
time==> 8.533699990948662 milliseconds
time==> 6.938199992873706 milliseconds
time==> 7.728199998382479 milliseconds
time==> 8.973399992100894 milliseconds
time==> 7.097900001099333 milliseconds
time==> 7.361600000876933 milliseconds
time==> 7.19540000136476 milliseconds
time==> 6.9023000105516985 milliseconds
time==> 6.689100002404302 milliseconds
time==> 6.8738999980269 milliseconds
time==> 6.7260999931022525 milliseconds
```

Fig. 8: Execution time of XGboost Efficiency Algorithm based on experiment-II

Table 1: Comparison of Different Algorithms

Algorithm	Accuracy(percentage)	Efficiency (sec)
Binary Decision Tree	99	0.001
XGBoost	99	0.008
SVM	96	0.001

5. Conclusion

We have implemented three machine learning algorithms to identify the DDoS attack which includes Binary decision tree, XGBoost and SVM. From the result SVM is giving very less accuracy which is 96 percentage and as per security concern we can't compromise the network environment for security. XGboost and binary decision tree algorithms are giving same accuracy rate which is 99 percentage and that is actually very effective. But when we compared the efficiency of this two algorithm the Binary decision tree is taking less time compared to XGboost. For experiment -I prediction time of Binary decision tree is taking 0.001 second and XGBoost is taking 0.008 second. When we talk about bulk of data this difference is very huge. So Binary Decision tree is more effective in terms of accuracy and efficient algorithm in the line of speed of detection for identifying the DDoS attack. We are also using standard methods of raising alarms in multiple levels. In future work fine grain classification of various DDos attacks is planned to be done for better resolution ad appropriate sending up of alarms.

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